

Berry curvature and response functions from VASP wavefunctions

VASPBERRY 1.3 scientific examples

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Abstract

VASPBERRY evaluates geometric and response properties from VASP electronic states. This report illustrates the discrete-wavefunction and Kubo approaches with monolayer MoS₂ and a Bi bilayer. The examples cover reciprocal-space Berry curvature, Chern and Z_2 indices, intrinsic charge Hall response, circular optical transitions and real-space spinor densities. Figures use physical Cartesian coordinates or distances along named high-symmetry paths; the associated calculation guides provide the VASP files and executable commands.

1. Methods

1.1 Berry curvature from wavefunction overlaps

The Fukui–Hatsugai–Suzuki method evaluates the Berry phase around each cell of a periodic $\mathbf{k}$ mesh using overlaps between neighboring wavefunctions. For an isolated group of bands, the overlap is a matrix and its determinant gives a gauge-invariant loop phase. With the connection $\mathbf{A} = i\langle u | \nabla_{\mathbf{k}} u \rangle$ used here,

$$\Phi_p = -\arg(U_1(\mathbf{k})U_2(\mathbf{k} + \Delta\mathbf{k}_1)U_1(\mathbf{k} + \Delta\mathbf{k}_2)^{-1}U_2(\mathbf{k})^{-1}), \quad \overline{\Omega}_p = \frac{\Phi_p}{\Delta S_k}.$$

The normalized link U_j is the phase of the overlap determinant. $\Delta S_k = |\mathbf{b}_1 \times \mathbf{b}_2|/(N_1 N_2)$ is the area of a mesh cell, with reciprocal vectors containing 2π . Thus Φ_p is a phase and $\overline{\Omega}_p$ is an area-averaged curvature in $\text{\AA}^2$. The Chern number is $C = (2\pi)^{-1} \sum_p \Phi_p$. An isolated occupied group can include internal degeneracies; its separation from excluded bands remains essential. The discrete construction and its band-group extension are described by Fukui, Hatsugai and Suzuki.

The map and integral convey different information. A material can have strong local curvature at opposite valleys while its total Chern number is zero. An integer lattice Chern result also needs a sufficiently resolved mesh and a physically appropriate isolated band group.

1.2 Kubo curvature and Hall conductivity

For an isolated band and a Cartesian Hamiltonian derivative $D_\alpha = \partial H / \partial k_\alpha$,

$$\Omega_{n,xy}(\mathbf{k}) = -2 \text{Im} \sum_{m \neq n} \frac{D_{x,nm} D_{y,mn}}{(E_n - E_m)^2}.$$

This form resolves curvature at each sampled $\mathbf{k}$ point. Sharp curvature near small gaps makes the $\mathbf{k}$ sampling and intermediate-band window particularly important. Individual-band values are not used where the sampled bands cannot be resolved separately. The native WAVECAR implementation uses canonical momentum of the stored pseudo-wavefunctions; full material velocity can require PAW, nonlocal and SOC terms. The exported-matrix interface permits an explicitly supplied physical operator.

For a two-dimensional system, the intrinsic charge sheet conductivity is

$$\sigma_{xy}(\mu, T) = -\frac{e^2}{h} \sum_n \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} f(E_{n\mathbf{k}} - \mu, T) \Omega_{n,xy}(\mathbf{k}).$$

Within an insulating gap at zero temperature, this becomes $\sigma_{xy} = -(e^2/h)C_{\text{occ}}$ in the stated convention. General background on Berry curvature and electronic transport is given in Xiao, Chang and Niu. The gap-only Bi example below evaluates the occupied-group flux directly; it does not assign independent curvatures to unresolved Kramers partners.

2. Materials and numerical settings

Dataset	Wavefunctions and sampling	Quantities shown
Monolayer MoS ₂ , full zone	SOC; 12×12 mesh; 400 eV; occupied bands 1–18	Fukui curvature map and Chern integral
Monolayer MoS ₂ , band path	SOC; 48 K–Γ–K′ points; 32 bands; 400 eV	Kubo curvature, optical spectra, Γ-point state density
Buckled Bi bilayer	SOC; full 12×12 mesh; 18 bands; occupied bands 1–10	Chern number, Z_2 and insulating-gap charge Hall response

The input guide links the structures and VASP files. The path and full-zone datasets are separate calculations. For the full-zone MoS₂ example, VASP 5.4.4 regenerated the 26-band SOC wavefunctions on a full 12×12 mesh from the public SCF charge density, with ICHARG=11 and ISYM=−1. The sampled occupied-to-empty direct gap is 1.674 eV. The tutorial supplies the preparation files and commands to generate this 149 MB WAVECAR locally. The other examples use supplied or downloadable WAVECAR files.

3. Results

3.1 MoS₂: Fukui Berry curvature over the full Brillouin zone

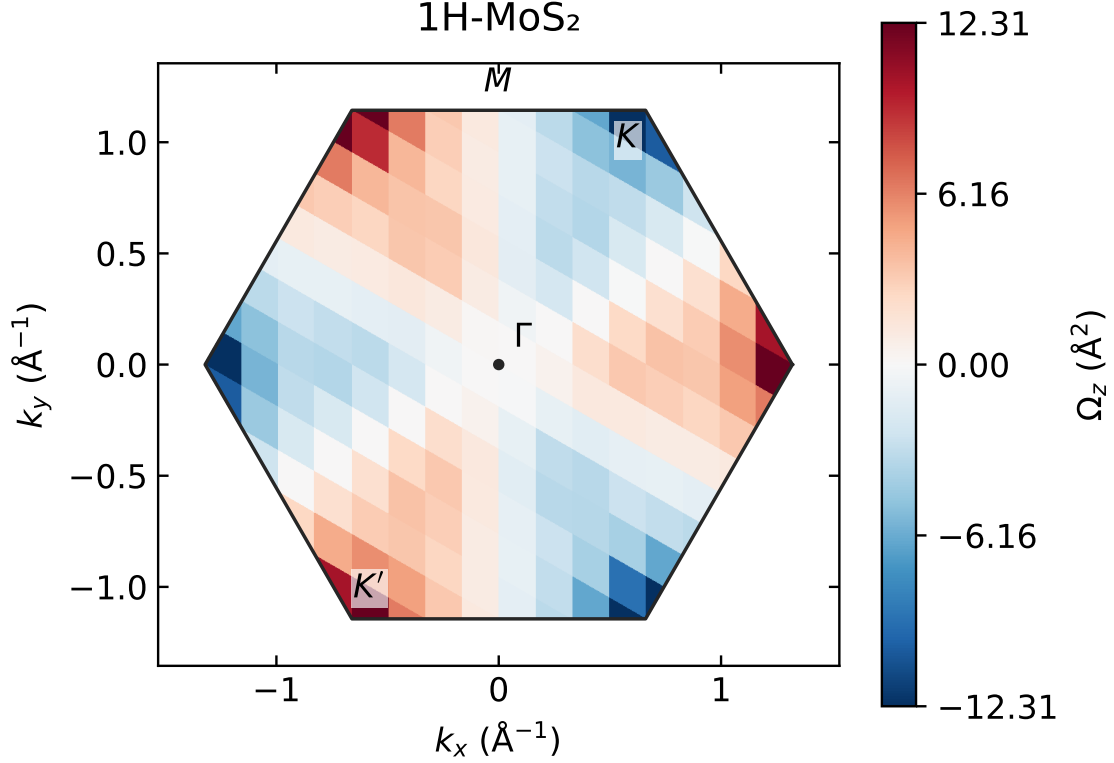


Figure 1. Occupied-group Berry curvature of monolayer MoS₂ in Cartesian k_x, k_y coordinates ($\text{\AA}^{-1}$). The boundary is the hexagonal first Brillouin zone, with Γ , K , K' and M identified. Colors represent curvature in $\text{\AA}^2$; the plotted cells retain the 12×12 sampling without interpolation. The source is a new full-zone VASP calculation followed by native VASPBERRY overlaps.

The curvature has opposite signs near the time-reversed K and K' valleys. The largest sampled magnitudes are approximately **12.31** $\text{\AA}^2$, while the full-zone Chern number is zero to numerical precision. The native text map integrates to approximately -5×10^{-7} after decimal rounding. This illustrates the distinction between valley-contrasting local geometry and vanishing total charge Chern number. The connection between inversion breaking, spin-orbit coupling and valley optical response in MoS₂ is discussed by Xiao et al.. The present map is a finite-mesh example rather than a convergence study.

Calculation and plotting commands

3.2 MoS₂: dispersion and Kubo curvature along K– Γ –K'

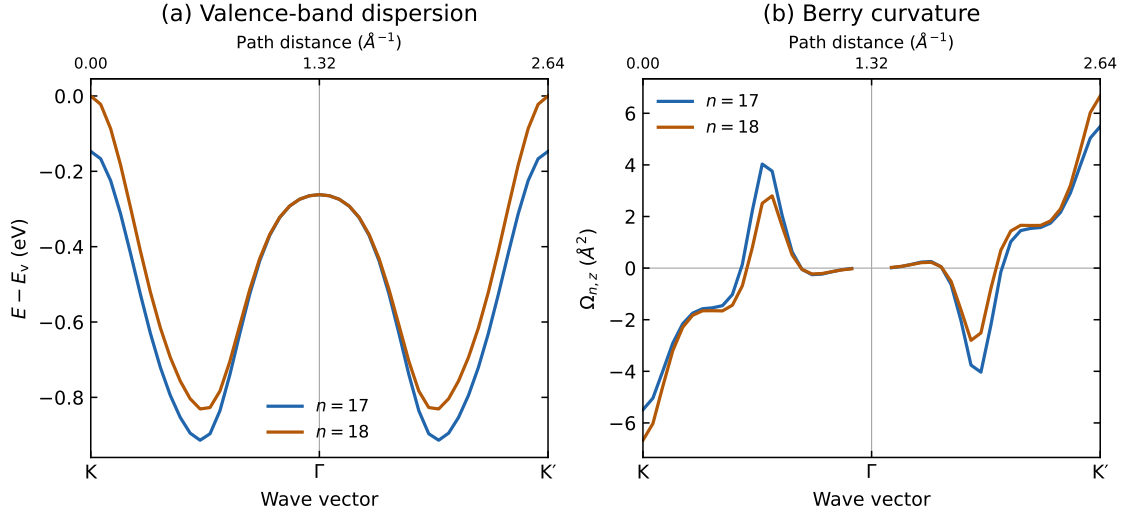


Figure 2. Valence-band dispersion and canonical-momentum Kubo curvature of bands 17 and 18 along K– Γ –K'. The horizontal positions follow Cartesian path distance. Gaps in the curvature curves indicate unresolved individual states near Γ ; those values are not connected through the degeneracy.

Band 18 has opposite curvature at the two valleys, approximately $-6.67 \text{ \AA}^2$ at K and $+6.67 \text{ \AA}^2$ at K' for this band window. These values are pointwise, band-resolved quantities. Figure 1 instead shows the finite-cell curvature of the complete occupied group, so the two figures are not a point-for-point comparison of identical observables. A band path also cannot supply the area integral needed for a Chern number or Hall conductivity.

Calculation and interpretation

3.3 MoS₂: valley-selective circular transitions

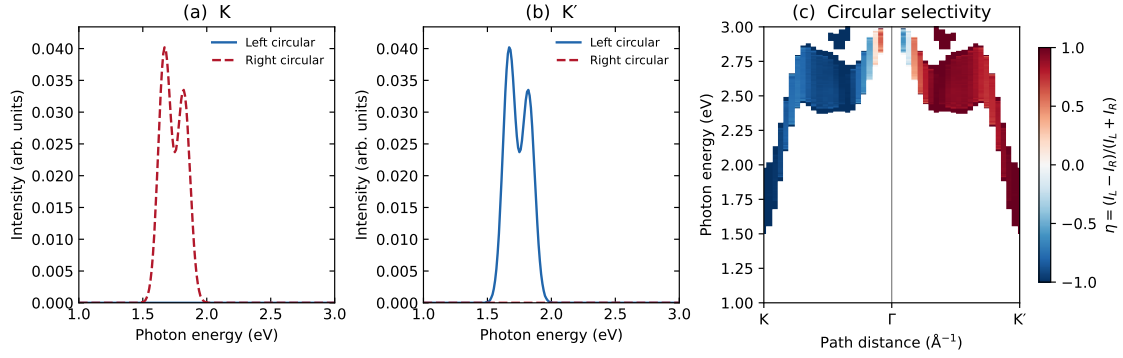


Figure 3. Left- and right-circular spectra at K and K' and the selectivity $\eta = (I_L - I_R)/(I_L + I_R)$ along the path. Normal incidence is used, with 0.05 eV Gaussian broadening and transitions from occupied bands 1–18 to bands 19–20. White regions in the selectivity map exclude negligible intensity.

The first peak occurs near **1.67 eV**. Its selectivity is -1 at K and $+1$ at K' in VASPBERRY's channel convention. The opposite helicities provide a clear illustration of valley-selective transitions. Intensities are in arbitrary units and retain the native sampling normalization; an absolute absorption spectrum requires a full-zone calculation and the appropriate optical matrix elements.

Optical calculation guide

3.4 Bi: Chern number, Z_2 and charge Hall response

The complete occupied group of the Bi bilayer gives $\mathbf{C} = \mathbf{0}$ and $Z_2 = 1$. These results are compatible: the Z_2 invariant diagnoses a time-reversal-symmetric topological phase even when the charge Chern number vanishes. The sampled minimum direct gap is about **0.592 eV**, and the global band gap is about **0.510 eV**.

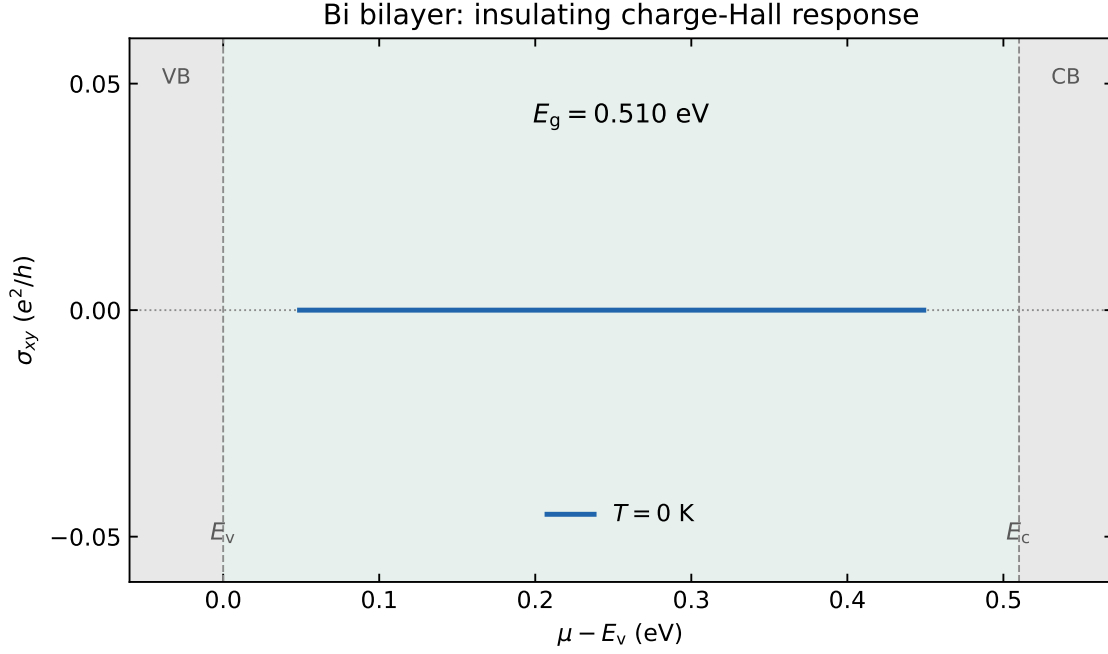


Figure 4. Intrinsic charge sheet conductivity at zero temperature across the sampled Bi insulating gap. The occupied-group Fukui result is zero throughout the calculated chemical-potential interval. Band-edge markers show the available gap; no metallic response is inferred beyond the scan.

The result is the expected charge-Hall cancellation for this time-reversal symmetric system. It does not represent a nonzero valley-Hall signal. The Chern guide and Z_2 guide provide the corresponding full-zone figures. The Z_2 n-field is gauge dependent; its parity, rather than its local color pattern, is the reported topological invariant.

Hall calculation guide

3.5 MoS₂: a real-space Γ state

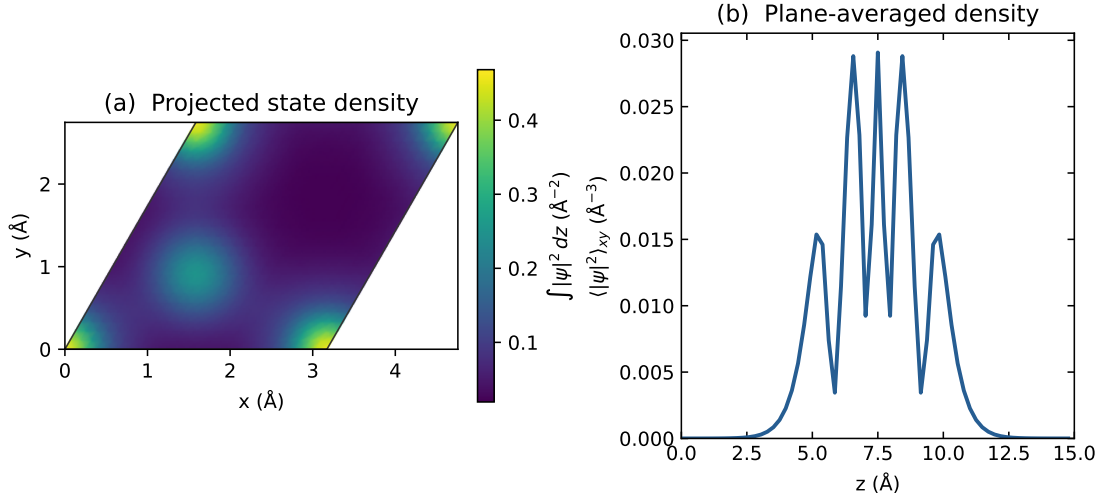


Figure 5. Projected pseudo-wavefunction density of band 18 at Γ and its plane average along z . The in-plane plot uses the Cartesian geometry of the oblique primitive cell. Both spinor components are retained.

The integrated pseudo-density is approximately **0.8258**, consistent with the plane-wave coefficient norm. The PAW augmentation is not part of this plot, so a unit density integral is not imposed. The example illustrates conversion of complex amplitudes into a physical-coordinate density map.

Wavefunction calculation guide

4. Practical use

Begin with the example guide, reproduce the chosen quantity, and then select the sampling and band window appropriate to your own material. Use Cartesian reciprocal-space maps to display lattice symmetry, named k paths for band-resolved quantities, and energy or chemical potential for spectra and transport. State units, represented bands, temperature and broadening in the caption.

For quantitative conclusions, converge the VASP electronic structure, k mesh and relevant band window. The examples show how the methods are used; the supplementary analytic references check formulas and conventions, while validation details document the numerical implementation. Detailed execution records remain alongside the numerical files for reproducibility.

References

1. T. Fukui, Y. Hatsugai and H. Suzuki, *Chern Numbers in Discretized Brillouin Zone: Efficient Method of Computing (Spin) Hall Conductances*, J. Phys. Soc. Jpn. **74**, 1674–1677 (2005).
2. D. Xiao, M.-C. Chang and Q. Niu, *Berry Phase Effects on Electronic Properties*, Rev. Mod. Phys. **82**, 1959–2007 (2010).
3. D. Xiao, G.-B. Liu, W. Feng, X. Xu and W. Yao, *Coupled Spin and Valley Physics in Monolayers of MoS₂ and Other Group-VI Dichalcogenides*, Phys. Rev. Lett. **108**, 196802 (2012).